

# Generalised Linear Models

Logistic regression

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## Q: Survival of passengers on the Titanic ~ Class

Read `titanic_long.csv` dataset and fit linear model (survival ~ class).

|   | class | age   | sex  | survived |
|---|-------|-------|------|----------|
| 1 | first | adult | male | 1        |
| 2 | first | adult | male | 1        |
| 3 | first | adult | male | 1        |
| 4 | first | adult | male | 1        |
| 5 | first | adult | male | 1        |
| 6 | first | adult | male | 1        |

## Quiz: Did passenger class influence survival?

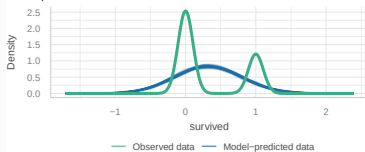
<https://pollev.com/franciscorod726>

# Let's check linear model:

```
m5 <- lm(survived ~ class, data = titanic)
library('easystats')
check_model(m5)
```

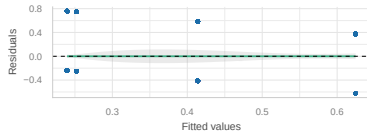
## Posterior Predictive Check

Model-predicted lines should resemble observed data line



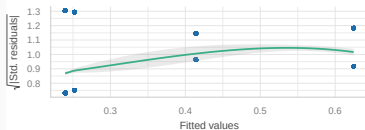
## Linearity

Reference line should be flat and horizontal



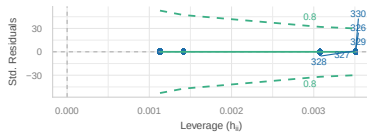
## Homogeneity of Variance

Reference line should be flat and horizontal



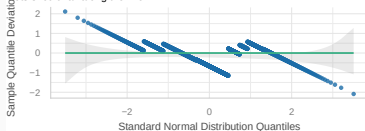
## Influential Observations

Points should be inside the contour lines

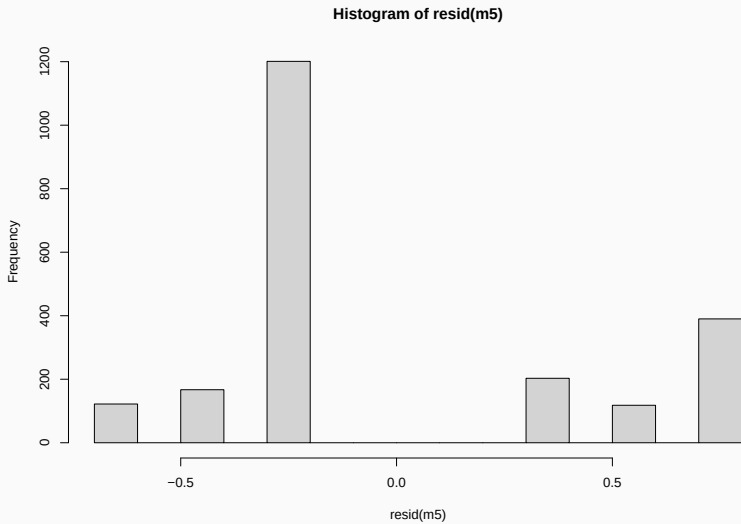


## Normality of Residuals

Points should fall along the line



## Weird residuals!



What if your residuals are clearly non-normal  
or variance not constant (heteroscedasticity)?

- Binary variables (0/1)

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- **Generalised Linear Models to the rescue!**

1. Response variable - distribution family

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- Bernoulli - Binomial

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- etc

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## 2. **Predictors** (continuous or categorical)

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## 2. Predictors (continuous or categorical)

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- Gaussian: identity
- Binomial: logit, probit
- Poisson: log...
- See **family**.

# The modelling process

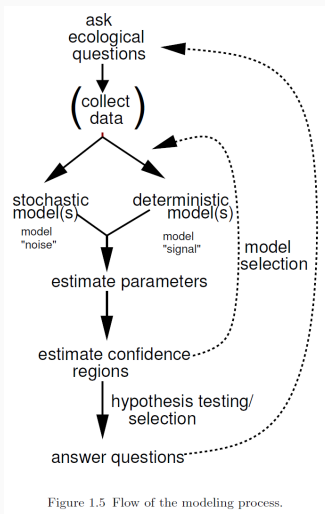


Figure 1.5 Flow of the modeling process.

## Bernoulli - Binomial distribution (Logistic regression)

Response variable: **Yes/No** (e.g. survival, sex, presence/absence)

Canonical link function: **logit** (*log odds*), but others possible (see **family**)

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right)$$

Then

$$\text{logit}(P(\text{alive})) = a + bx$$

$$P(\text{alive}) = \text{invlogit}(a + bx) = \frac{e^{a+bx}}{1 + e^{a+bx}}$$

## Where is the variance?

In a Gaussian GLM

$$y \sim \text{Normal}(\mu, \sigma)$$

In a Binomial GLM

$$y \sim \text{Binomial}(n, p)$$

**n** = number of trials

**p** = probability of success

$$\text{Var}(y) = np(1 - p)$$

(maximum variance when **p** around 0.5)

## Back to survival of Titanic passengers

---

## How many survived in each class?

```
table(titanic$class, titanic$survived)
```

|        | 0   | 1   |
|--------|-----|-----|
| crew   | 673 | 212 |
| first  | 122 | 203 |
| second | 167 | 118 |
| third  | 528 | 178 |

## How many survived in each class? (*dplyr*)

```
titanic |>  
  group_by(class, survived) |>  
  summarise(count = n())
```

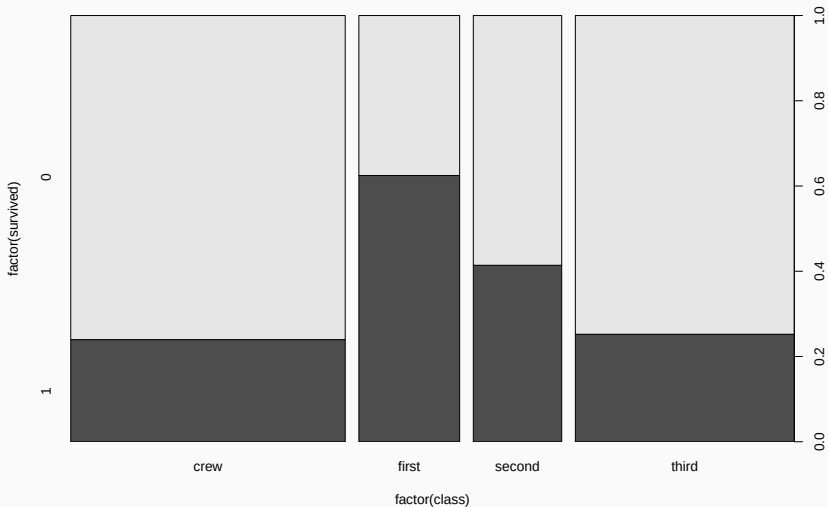
```
# A tibble: 8 x 3
```

```
# Groups:   class [4]
```

|   | class  | survived | count |
|---|--------|----------|-------|
|   | <chr>  | <int>    | <int> |
| 1 | crew   | 0        | 673   |
| 2 | crew   | 1        | 212   |
| 3 | first  | 0        | 122   |
| 4 | first  | 1        | 203   |
| 5 | second | 0        | 167   |
| 6 | second | 1        | 118   |
| 7 | third  | 0        | 528   |
| 8 | third  | 1        | 178   |

## Data visualisation (mosaic plot)

```
plot(factor(survived) ~ factor(class), data = titanic)
```



```
tit.glm <- glm(survived ~ class,  
               data = titanic,  
               family = binomial)
```

which corresponds to

$$\text{logit}(P(\text{survival})_i) = a + b \cdot \text{class}_i$$

$$\text{logit}(P(\text{survival})_i) = a + b_{\text{first}} + c_{\text{second}} + d_{\text{third}}$$

# Interpreting binomial GLM

```
tit.glm <- glm(survived ~ class, data = titanic, family = binomial)
```

Call:

```
glm(formula = survived ~ class, family = binomial, data = titanic)
```

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | -1.15516 | 0.07876    | -14.667 | < 2e-16 ***  |
| classfirst  | 1.66434  | 0.13902    | 11.972  | < 2e-16 ***  |
| classecond  | 0.80785  | 0.14375    | 5.620   | 1.91e-08 *** |
| classtthird | 0.06785  | 0.11711    | 0.579   | 0.562        |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 2769.5 on 2200 degrees of freedom  
Residual deviance: 2588.6 on 2197 degrees of freedom  
AIC: 2596.6

Number of Fisher Scoring iterations: 4

## Binomial GLM estimates are in `logit` scale!

We need to **back-transform** (apply *inverse logit*):

- Manually: `plogis`

## Binomial GLM estimates are in `logit` scale!

We need to **back-transform** (apply *inverse logit*):

- Manually: `plogis`
- Automatically: `easystats`, etc.

## Interpreting logistic regression output (easystats)

```
library('easystats') # 'modelbased' pkg  
estimate_means(tit.glm)
```

Estimated Marginal Means

| class  | Probability | 95% CI       |
|--------|-------------|--------------|
| crew   | 0.24        | [0.21, 0.27] |
| first  | 0.62        | [0.57, 0.68] |
| second | 0.41        | [0.36, 0.47] |
| third  | 0.25        | [0.22, 0.29] |

Variable predicted: survived

Predictors modulated: class

Predictions are on the response-scale.

# Analysing differences among factor levels (class)

```
estimate_contrasts(tit.glm)
```

## Marginal Contrasts Analysis

| Level1 | Level2 | Difference | SE   | 95% CI         | z      | p      |
|--------|--------|------------|------|----------------|--------|--------|
| -----  |        |            |      |                |        |        |
| first  | crew   | 0.39       | 0.03 | [ 0.33, 0.44]  | 12.65  | < .001 |
| second | crew   | 0.17       | 0.03 | [ 0.11, 0.24]  | 5.37   | < .001 |
| third  | crew   | 0.01       | 0.02 | [-0.03, 0.06]  | 0.58   | 0.563  |
| second | first  | -0.21      | 0.04 | [-0.29, -0.13] | -5.31  | < .001 |
| third  | first  | -0.37      | 0.03 | [-0.43, -0.31] | -11.85 | < .001 |
| third  | second | -0.16      | 0.03 | [-0.23, -0.10] | -4.84  | < .001 |

Variable predicted: survived

Predictors contrasted: class

p-values are uncorrected.

Contrasts are on the response-scale.

```
library('easystats')    # 'performance' pkg  
r2(tit.glm)
```

```
# R2 for Logistic Regression  
Tjur's R2: 0.087
```

But there are caveats (e.g. see [here](#) and [here](#))

```
kable(xtable::xtable(tit.glm), digits = 2)
```

|             | Estimate | Std. Error | z value | Pr(> z ) |
|-------------|----------|------------|---------|----------|
| (Intercept) | -1.16    | 0.08       | -14.67  | 0.00     |
| classfirst  | 1.66     | 0.14       | 11.97   | 0.00     |
| classecond  | 0.81     | 0.14       | 5.62    | 0.00     |
| classtthird | 0.07     | 0.12       | 0.58    | 0.56     |

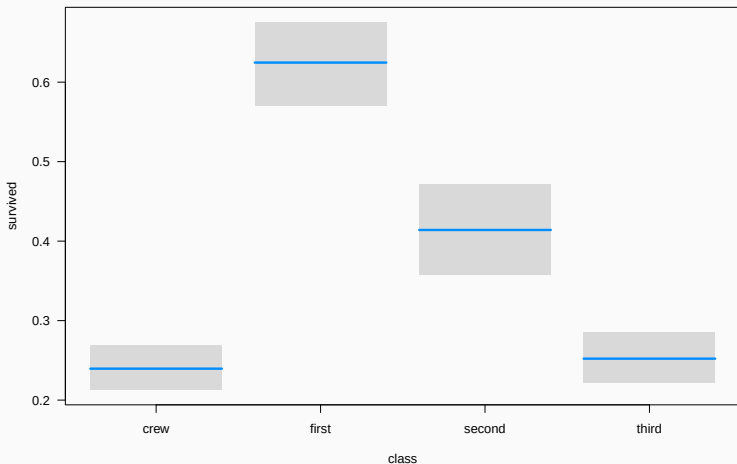
## Presenting model results

```
library('modelsummary')  
modelsummary(tit.glm, output = 'markdown')
```

|             | (1)               |
|-------------|-------------------|
| (Intercept) | -1.155<br>(0.079) |
| classfirst  | 1.664<br>(0.139)  |
| classecond  | 0.808<br>(0.144)  |
| classtthird | 0.068<br>(0.117)  |
| Num.Obs.    | 2201              |
| AIC         | 2596.6            |
| BIC         | 2619.3            |
| Log.Lik.    | -1294.278         |
| F           | 57.742            |

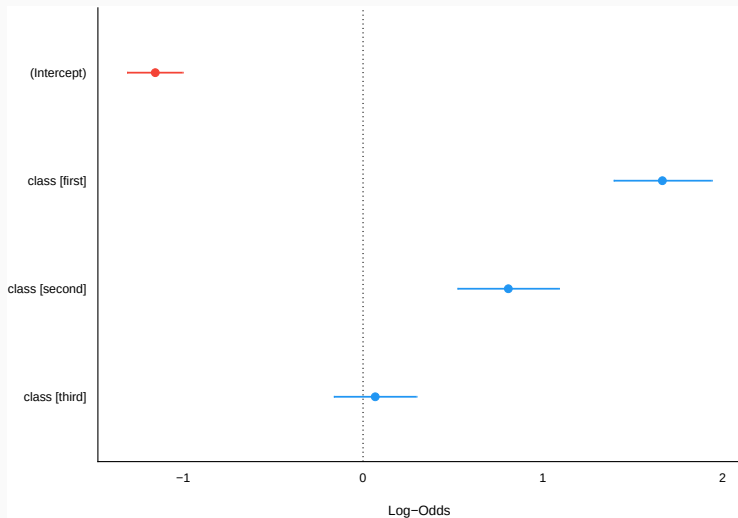
## Visualising model: visreg package

```
visreg(tit.glm, scale = 'response', rug = FALSE)
```



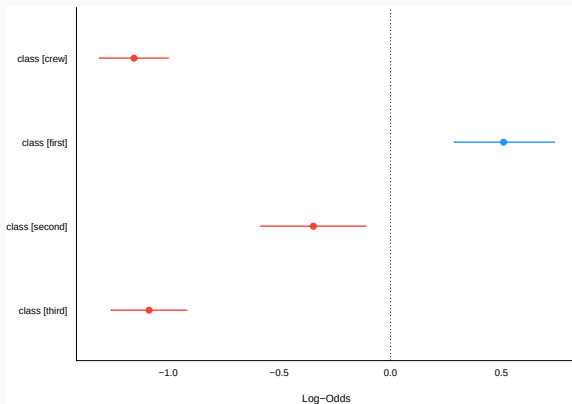
## Visualising model: easystats (see package)

```
plot(parameters(tit.glm), show_intercept = TRUE)
```



## Model without intercept

```
no.intercept <- glm(survived ~ class - 1, family = binomial, data =  
plot(parameters(no.intercept))
```

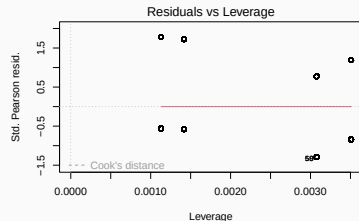
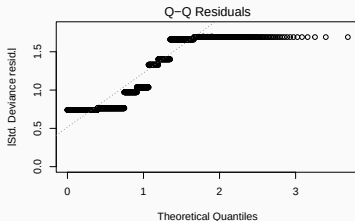
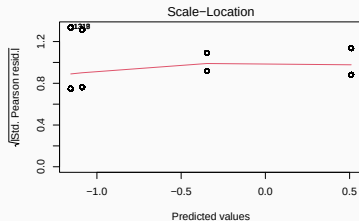
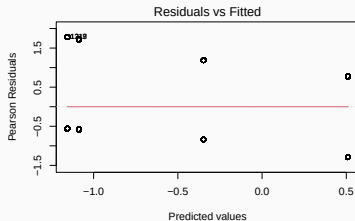


## Model checking

---

# `plot(model)` not very useful with binomial GLM

```
plot(tit.glm)
```



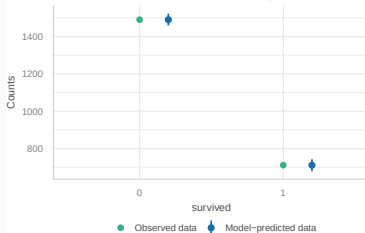
```
null device
```

# check\_model(easystats)

## check\_model(tit.glm)

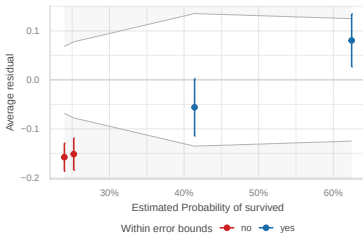
### Posterior Predictive Check

Model-predicted intervals should include observed data points



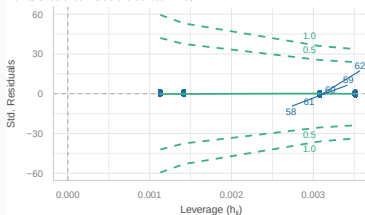
### Binned Residuals

Points should be within error bounds



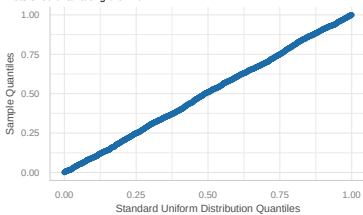
### Influential Observations

Points should be inside the contour lines



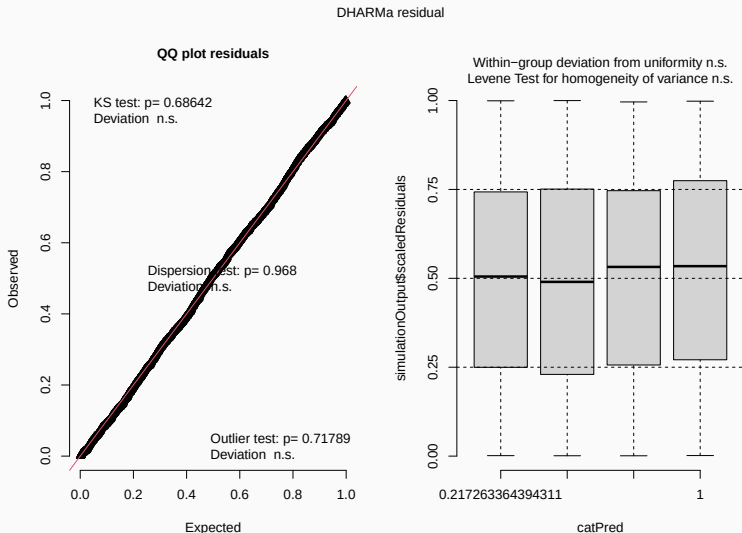
### Distribution of Quantile Residuals

Dots should fall along the line



# Residual diagnostics with DHARMa

```
library('DHARMa')  
simulateResiduals(tit.glm, plot = TRUE)
```

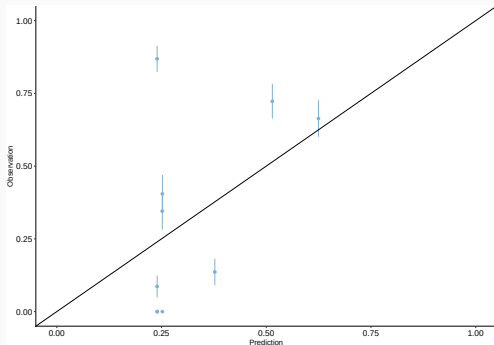


# Calibration plot

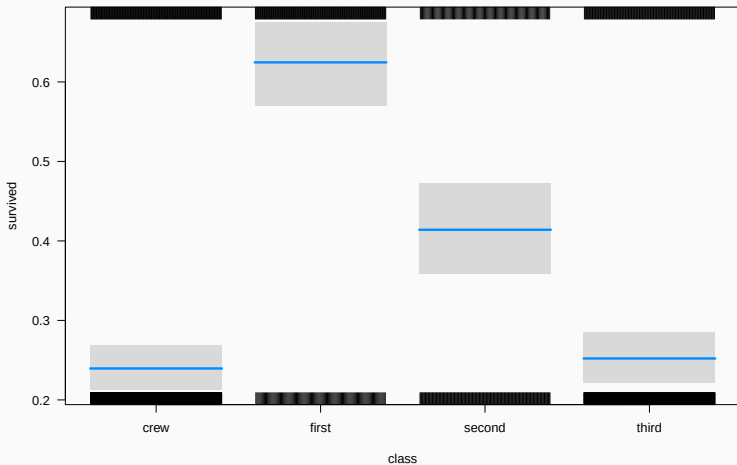
Compares predicted vs observed probabilities (grouped by quantiles)

```
library('predtools')
titanic$surv.pred <- predict(tit.glm, type = 'response')
calibration_plot(data = titanic, obs = 'survived', pred = 'surv.pred',
                 x_lim = c(0,1), y_lim = c(0,1))
```

\$calibration\_plot



Passenger class was important, but lots of unexplained variation



The goal is not to test whether the model's assumptions are 'true', because all models are false.

Rather, the goal is to assess exactly **how the model fails to describe the data**, as a path towards **model comprehension, revision, and improvement**.

Richard McElreath. *Statistical Rethinking*

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2. Fit model: `glm`. Don't forget to specify `family`!

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5. Plot model: `visreg`, ...

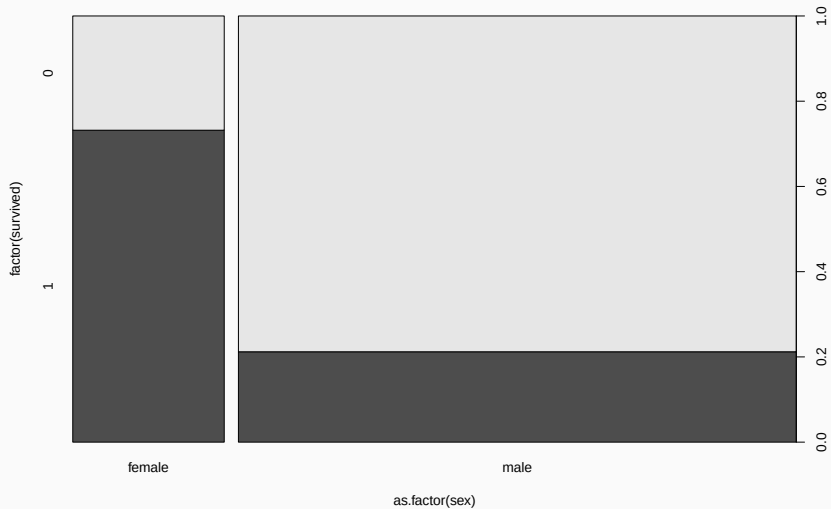
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4. Back-transform parameters from *logit* into probability scale  
(`estimate_means`)
5. Plot model: `visreg`, ...
6. Check model: `check_model`, `DHARMA::simulateResiduals`,  
`calibration_plot`

Q: Did men have higher survival  
than women?

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<https://pollev.com/franciscorod726>

## First, visualise data



Call:

```
glm(formula = survived ~ sex, family = binomial, data = titanic)
```

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z )   |
|-------------|----------|------------|---------|------------|
| (Intercept) | 1.0044   | 0.1041     | 9.645   | <2e-16 *** |
| sexmale     | -2.3172  | 0.1196     | -19.376 | <2e-16 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 2769.5 on 2200 degrees of freedom  
Residual deviance: 2335.0 on 2199 degrees of freedom  
AIC: 2339

Number of Fisher Scoring iterations: 4

# Model interpretation

```
estimate_means(tit.sex)
```

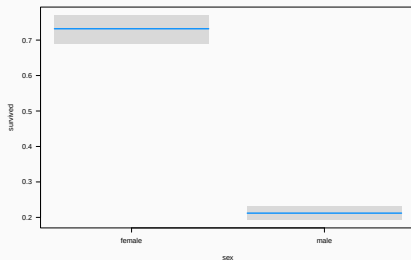
Estimated Marginal Means

| sex    | Probability | 95% CI       |
|--------|-------------|--------------|
| female | 0.73        | [0.69, 0.77] |
| male   | 0.21        | [0.19, 0.23] |

Variable predicted: survived

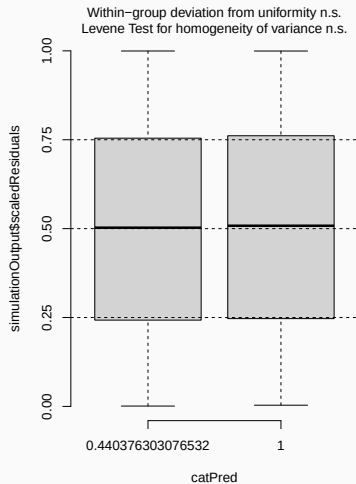
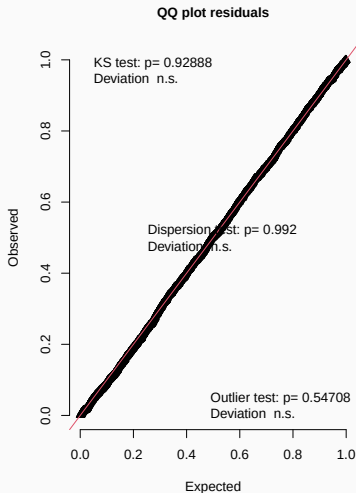
Predictors modulated: sex

Predictions are on the response-scale.



```
simulateResiduals(tit.sex, plot = TRUE)
```

DHARMA residual

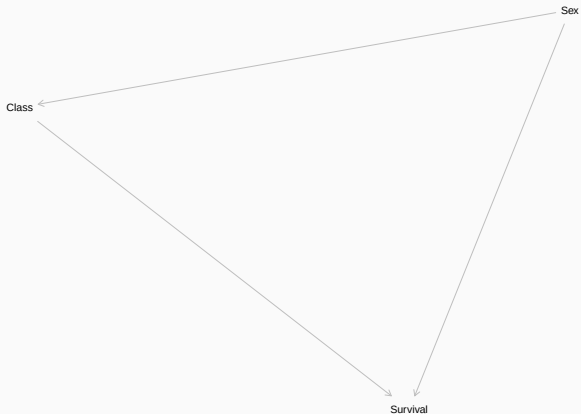


Q: Did women have higher survival because they travelled more in first class?

---

# Did women have higher survival because they travelled more in first class?

Sex is a confounder



## Let's look at the data

```
table(titanic$class, titanic$survived, titanic$sex)
```

```
, , = female
```

|        | 0   | 1   |
|--------|-----|-----|
| crew   | 3   | 20  |
| first  | 4   | 141 |
| second | 13  | 93  |
| third  | 106 | 90  |

```
, , = male
```

|        | 0   | 1   |
|--------|-----|-----|
| crew   | 670 | 192 |
| first  | 118 | 62  |
| second | 154 | 25  |
| third  | 422 | 88  |

<https://pollev.com/franciscorod726>

## Fit additive model with both factors

Call:

```
glm(formula = survived ~ class + sex, family = binomial, data = titanic)
```

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z ) |     |
|-------------|----------|------------|---------|----------|-----|
| (Intercept) | 1.18740  | 0.15747    | 7.541   | 4.68e-14 | *** |
| classfirst  | 0.88081  | 0.15697    | 5.611   | 2.01e-08 | *** |
| classecond  | -0.07178 | 0.17093    | -0.420  | 0.675    |     |
| classtthird | -0.77742 | 0.14231    | -5.463  | 4.69e-08 | *** |
| sexmale     | -2.42133 | 0.13909    | -17.408 | < 2e-16  | *** |
| ---         |          |            |         |          |     |

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

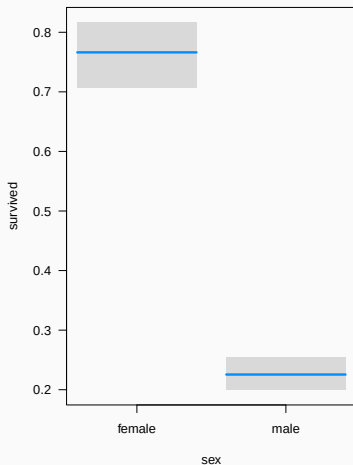
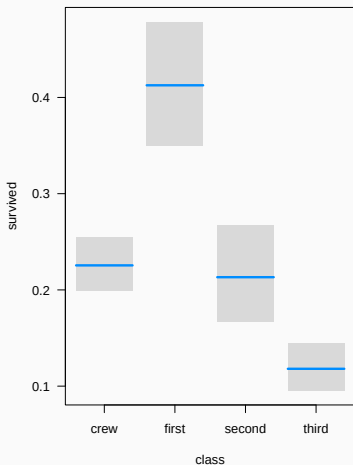
(Dispersion parameter for binomial family taken to be 1)

Null deviance: 2769.5 on 2200 degrees of freedom  
Residual deviance: 2228.9 on 2196 degrees of freedom  
AIC: 2238.9

Number of Fisher Scoring iterations: 4

## Plot additive model

```
visreg(tit.sex.class.add, scale = 'response', rug = FALSE)
```



null device

## Fit model with the interaction of both factors

Call:

```
glm(formula = survived ~ class * sex, family = binomial, data = titanic)
```

Coefficients:

|                     | Estimate | Std. Error | z value | Pr(> z ) |     |
|---------------------|----------|------------|---------|----------|-----|
| (Intercept)         | 1.89712  | 0.61914    | 3.064   | 0.00218  | **  |
| classfirst          | 1.66535  | 0.80026    | 2.081   | 0.03743  | *   |
| classecond          | 0.07053  | 0.68630    | 0.103   | 0.91815  |     |
| classtthird         | -2.06075 | 0.63551    | -3.243  | 0.00118  | **  |
| sexmale             | -3.14690 | 0.62453    | -5.039  | 4.68e-07 | *** |
| classfirst:sexmale  | -1.05911 | 0.81959    | -1.292  | 0.19627  |     |
| classecond:sexmale  | -0.63882 | 0.72402    | -0.882  | 0.37760  |     |
| classtthird:sexmale | 1.74286  | 0.65139    | 2.676   | 0.00746  | **  |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 2769.5 on 2200 degrees of freedom

Residual deviance: 2163.7 on 2193 degrees of freedom

AIC: 2179.7

## Women had higher survival than men, even within the same class

```
estimate_means(tit.sex.class.int)
```

Estimated Marginal Means

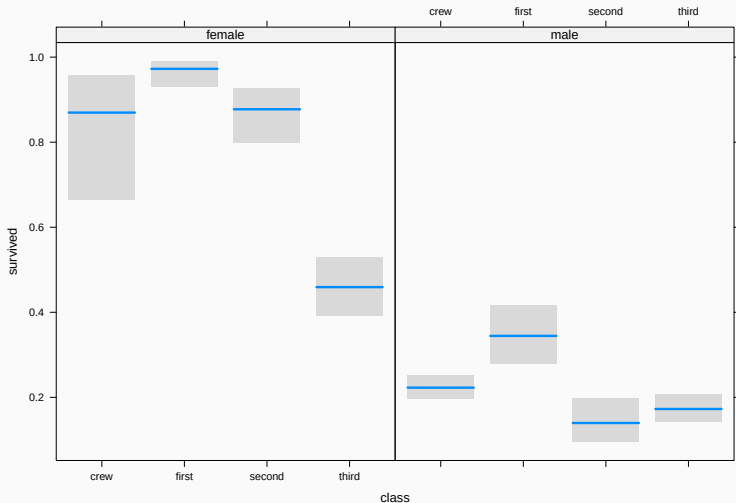
| class  | sex    | Probability | 95% CI       |
|--------|--------|-------------|--------------|
| -----  |        |             |              |
| crew   | female | 0.87        | [0.66, 0.96] |
| first  | female | 0.97        | [0.93, 0.99] |
| second | female | 0.88        | [0.80, 0.93] |
| third  | female | 0.46        | [0.39, 0.53] |
| crew   | male   | 0.22        | [0.20, 0.25] |
| first  | male   | 0.34        | [0.28, 0.42] |
| second | male   | 0.14        | [0.10, 0.20] |
| third  | male   | 0.17        | [0.14, 0.21] |

Variable predicted: survived

Predictors modulated: class, sex

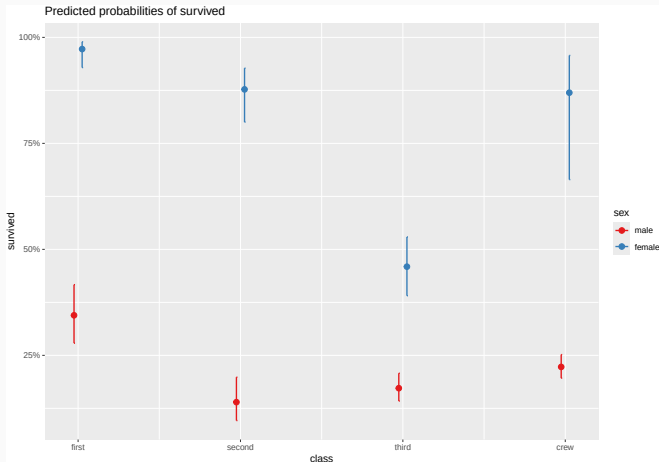
## Women had higher survival than men, even within the same class

```
visreg(tit.sex.class.int, by = 'sex', xvar = 'class', scale = 'response', rug = FALSE)
```



# Visualising model (sjPlot)

```
library('sjPlot')  
plot_model(tit.sex.class.int, type = 'int')
```



# Comparing models

```
library('easystats') # 'performance' pkg
compare_performance(tit.sex.class.add, tit.sex.class.int)
```

# Comparison of Model Performance Indices

| Name              | Model | AIC (weights)  | AICc (weights) | BIC (weights)  |
|-------------------|-------|----------------|----------------|----------------|
| tit.sex.class.add | glm   | 2238.9 (<.001) | 2238.9 (<.001) | 2267.4 (<.001) |
| tit.sex.class.int | glm   | 2179.7 (>.999) | 2179.8 (>.999) | 2225.3 (>.999) |

| Name              | Tjur's R2 | RMSE  | Sigma | Log_loss | Score_log |
|-------------------|-----------|-------|-------|----------|-----------|
| tit.sex.class.add | 0.248     | 0.405 | 1     | 0.506    | -Inf      |
| tit.sex.class.int | 0.271     | 0.399 | 1     | 0.492    | -Inf      |

| Name              | Score_spherical | PCP   |
|-------------------|-----------------|-------|
| tit.sex.class.add | 0.004           | 0.671 |
| tit.sex.class.int | 0.002           | 0.681 |

## Comparing parameters

```
compare_parameters(tit.sex.class.add, tit.sex.class.int)
```

| Parameter                   | tit.sex.class.add    | tit.sex.class.int    |
|-----------------------------|----------------------|----------------------|
| (Intercept)                 | 1.19 ( 0.88, 1.50)   | 1.90 ( 0.68, 3.11)   |
| class [first]               | 0.88 ( 0.57, 1.19)   | 1.67 ( 0.10, 3.23)   |
| class [second]              | -0.07 (-0.41, 0.26)  | 0.07 (-1.27, 1.42)   |
| class [third]               | -0.78 (-1.06, -0.50) | -2.06 (-3.31, -0.82) |
| sex [male]                  | -2.42 (-2.69, -2.15) | -3.15 (-4.37, -1.92) |
| class [first] × sex [male]  |                      | -1.06 (-2.67, 0.55)  |
| class [second] × sex [male] |                      | -0.64 (-2.06, 0.78)  |
| class [third] × sex [male]  |                      | 1.74 ( 0.47, 3.02)   |
| Observations                | 2201                 | 2201                 |

Is survival related to age?

Are age effects dependent on sex?

## Logistic regression for proportion data

---

## Why not modelling proportion?

- $1/4 = 0.25$

## Why not modelling proportion?

- $1/4 = 0.25$
- $250/1000 = 0.25$

## Why not modelling proportion?

- $1/4 = 0.25$
- $250/1000 = 0.25$
- Sample size information is lost!

## Read Titanic data in different format

Read `titanic_prop.csv` data.

|   | X | Class | Sex    | Age   | No  | Yes |
|---|---|-------|--------|-------|-----|-----|
| 1 | 1 | 1st   | Female | Adult | 4   | 140 |
| 2 | 2 | 1st   | Female | Child | 0   | 1   |
| 3 | 3 | 1st   | Male   | Adult | 118 | 57  |
| 4 | 4 | 1st   | Male   | Child | 0   | 5   |
| 5 | 5 | 2nd   | Female | Adult | 13  | 80  |
| 6 | 6 | 2nd   | Female | Child | 0   | 13  |

These are the same data, but summarized (see `Freq` variable).

## Use `cbind(n.success, n.failures)` as response

```
prop.glm <- glm(cbind(Yes, No) ~ Class, data = tit.prop, family = binomial)
```

Call:

```
glm(formula = cbind(Yes, No) ~ Class, family = binomial, data = tit.prop)
```

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | 0.5092   | 0.1146     | 4.445   | 8.79e-06 *** |
| Class2nd    | -0.8565  | 0.1661     | -5.157  | 2.51e-07 *** |
| Class3rd    | -1.5965  | 0.1436     | -11.114 | < 2e-16 ***  |
| ClassCrew   | -1.6643  | 0.1390     | -11.972 | < 2e-16 ***  |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 671.96 on 13 degrees of freedom  
Residual deviance: 491.06 on 10 degrees of freedom  
AIC: 545.68

Number of Fisher Scoring iterations: 4

## Survival probability by class

```
estimate_means(prop.glm)
```

Estimated Marginal Means

| Class | Probability | 95% CI       |
|-------|-------------|--------------|
| 1st   | 0.62        | [0.57, 0.68] |
| 2nd   | 0.41        | [0.36, 0.47] |
| 3rd   | 0.25        | [0.22, 0.29] |
| Crew  | 0.24        | [0.21, 0.27] |

Variable predicted: cbind(Yes, No)

Predictors modulated: Class

Predictors averaged: Yes (44), No (93)

Predictions are on the response-scale.

# Contrasts in survival among classes

## Marginal Contrasts Analysis

| Level1 | Level2 | Difference | SE   | 95% CI         | z      | p      |
|--------|--------|------------|------|----------------|--------|--------|
| 2nd    | 1st    | -0.21      | 0.04 | [-0.29, -0.13] | -5.31  | < .001 |
| 3rd    | 1st    | -0.37      | 0.03 | [-0.43, -0.31] | -11.85 | < .001 |
| Crew   | 1st    | -0.39      | 0.03 | [-0.44, -0.33] | -12.65 | < .001 |
| 3rd    | 2nd    | -0.16      | 0.03 | [-0.23, -0.10] | -4.84  | < .001 |
| Crew   | 2nd    | -0.17      | 0.03 | [-0.24, -0.11] | -5.37  | < .001 |
| Crew   | 3rd    | -0.01      | 0.02 | [-0.06, 0.03]  | -0.58  | 0.563  |

Variable predicted: cbind(Yes, No)

Predictors contrasted: Class

Predictors averaged: Yes (44), No (93)

p-values are uncorrected.

Contrasts are on the response-scale.

## Logistic regression with continuous predictors

---

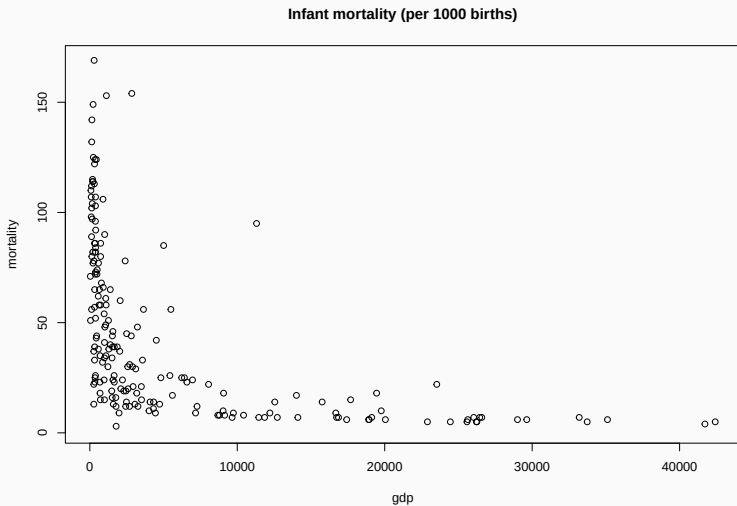
Example dataset: GDP and infant mortality

Read UN\_GDP\_infantmortality.csv.

| country          | mortality      | gdp           |
|------------------|----------------|---------------|
| Length:207       | Min. : 2.00    | Min. : 36     |
| Class :character | 1st Qu.: 12.00 | 1st Qu.: 442  |
| Mode :character  | Median : 30.00 | Median : 1779 |
|                  | Mean : 43.48   | Mean : 6262   |
|                  | 3rd Qu.: 66.00 | 3rd Qu.: 7272 |
|                  | Max. :169.00   | Max. :42416   |
|                  | NA's :6        | NA's :10      |

Q: Is infant mortality related to GDP?

<https://pollev.com/franciscorod726>



## Fit model

```
gdp.glm <- glm(cbind(mortality, 1000 - mortality) ~ gdp,  
               data = gdp, family = binomial)
```

Call:

```
glm(formula = cbind(mortality, 1000 - mortality) ~ gdp, family = binomial,  
    data = gdp)
```

Coefficients:

|             | Estimate   | Std. Error | z value | Pr(> z )   |
|-------------|------------|------------|---------|------------|
| (Intercept) | -2.657e+00 | 1.311e-02  | -202.76 | <2e-16 *** |
| gdp         | -1.279e-04 | 3.458e-06  | -36.98  | <2e-16 *** |

---

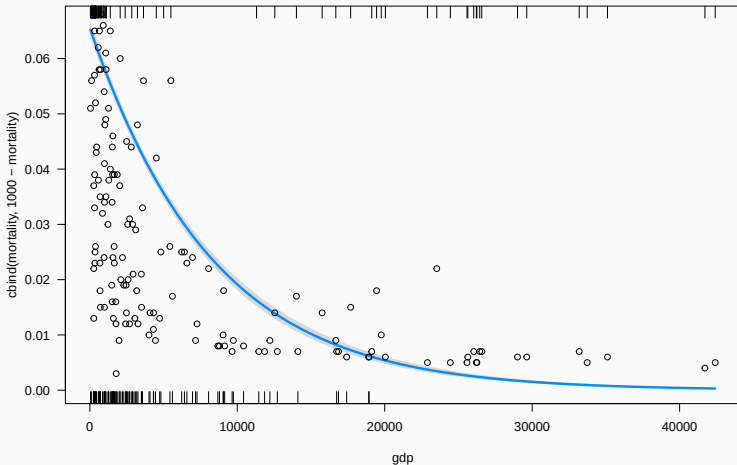
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 6430.2 on 192 degrees of freedom  
Residual deviance: 3530.2 on 191 degrees of freedom  
(14 observations deleted due to missingness)  
AIC: 4525.8

## Plot model using visreg:

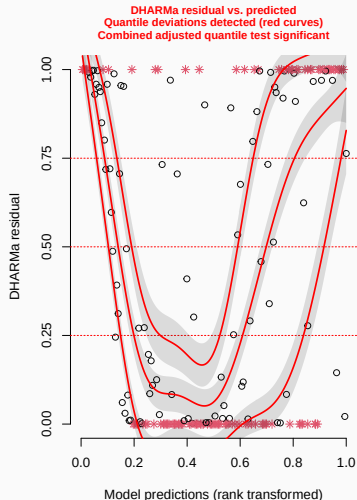
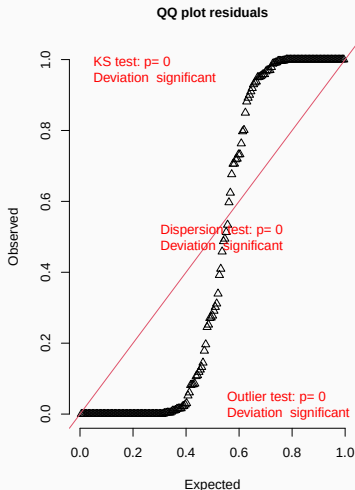
```
visreg(gdp.glm, scale = 'response')  
points(mortality/1000 ~ gdp, data = gdp)
```



# Residuals diagnostics with DHARMA

```
simulateResiduals(gdp.glm, plot = TRUE)
```

DHARMA residual



## Overdispersion

---

### Overdispersion:

more variation in the data than assumed by statistical model

$$\text{Var}(y) = np(1 - p)$$

## Testing for overdispersion (DHARMA)

```
simres <- simulateResiduals(gdp.glm, refit = TRUE)  
testDispersion(simres, plot = FALSE)
```

DHARMA nonparametric dispersion test via mean deviance residuals  
vs. simulated-refitted

```
data:  simres  
dispersion = 21, p-value < 2.2e-16  
alternative hypothesis: two.sided
```

```
check_overdispersion(gdp.glm)
```

```
# Overdispersion test
```

```
dispersion ratio = 2.933
```

```
p-value = < 0.001
```

`quasibinomial` allows us to model overdispersed binomial data

# Overdispersion in logistic regression with proportion data

```
gdp.overdisp <- glm(cbind(mortality, 1000 - mortality) ~ gdp,  
                    data = gdp, family = quasibinomial)
```

Call:

```
glm(formula = cbind(mortality, 1000 - mortality) ~ gdp, family = quasibinomial,  
    data = gdp)
```

Coefficients:

|             | Estimate   | Std. Error | t value | Pr(> t )     |
|-------------|------------|------------|---------|--------------|
| (Intercept) | -2.657e+00 | 5.977e-02  | -44.465 | < 2e-16 ***  |
| gdp         | -1.279e-04 | 1.577e-05  | -8.111  | 5.96e-14 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasibinomial family taken to be 20.7947)

Null deviance: 6430.2 on 192 degrees of freedom  
Residual deviance: 3530.2 on 191 degrees of freedom  
(14 observations deleted due to missingness)  
AIC: NA

# Mean estimates do not change after accounting for overdispersion

But standard errors (uncertainty) do!

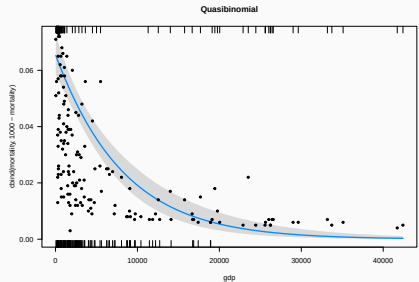
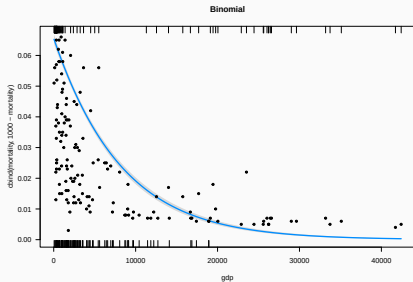
```
parameters(gdp.overdisp)
```

| Parameter   | Log-Odds  | SE       | 95% CI         | t(191) | p      |
|-------------|-----------|----------|----------------|--------|--------|
| (Intercept) | -2.66     | 0.06     | [-2.78, -2.54] | -44.46 | < .001 |
| gdp         | -1.28e-04 | 1.58e-05 | [ 0.00, 0.00]  | -8.11  | < .001 |

```
parameters(gdp.glm)
```

| Parameter   | Log-Odds  | SE       | 95% CI         | z       | p      |
|-------------|-----------|----------|----------------|---------|--------|
| (Intercept) | -2.66     | 0.01     | [-2.68, -2.63] | -202.76 | < .001 |
| gdp         | -1.28e-04 | 3.46e-06 | [ 0.00, 0.00]  | -36.99  | < .001 |

But standard errors (uncertainty) do!

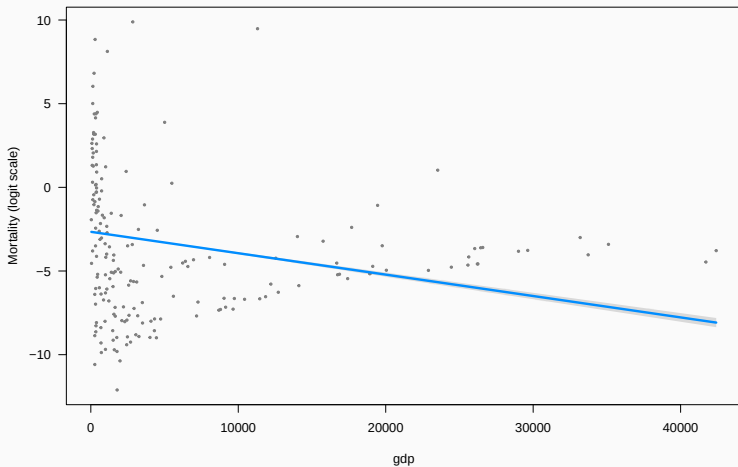


Think about the shape of  
relationships

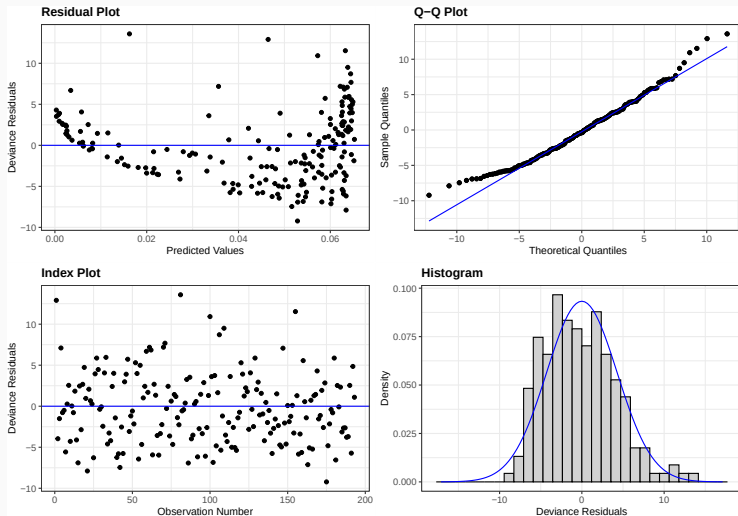
---

# Think about the shape of relationships

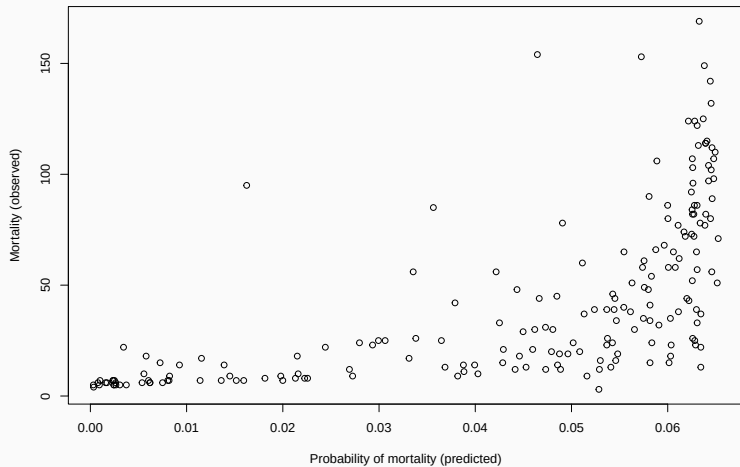
Not everything has to be linear...



# Residuals show non-linear pattern

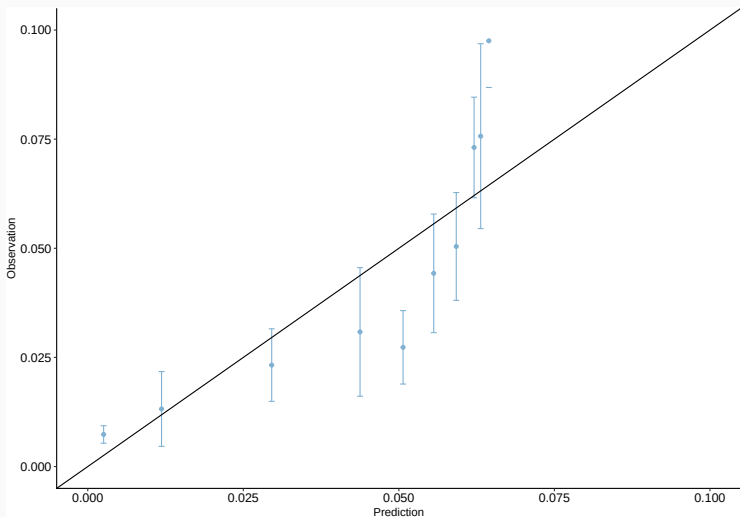


## Calibration plot shows non-linear pattern

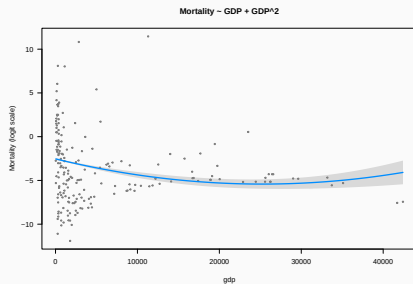
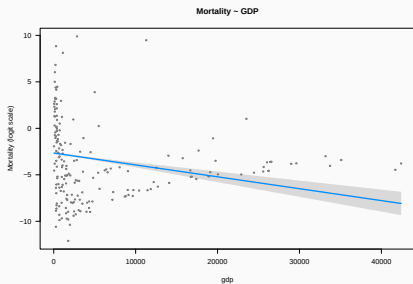


## Calibration plot shows non-linear pattern

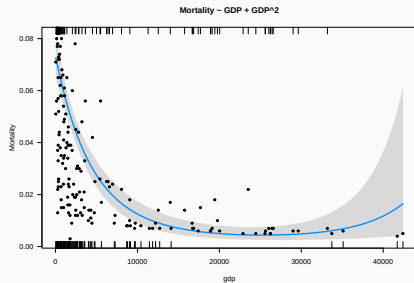
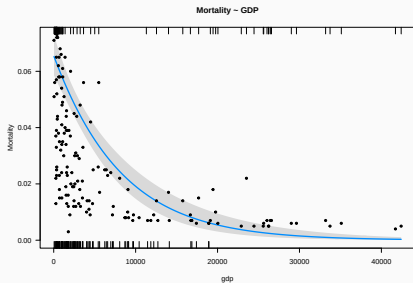
`$calibration_plot`



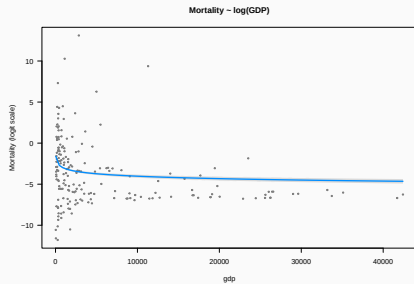
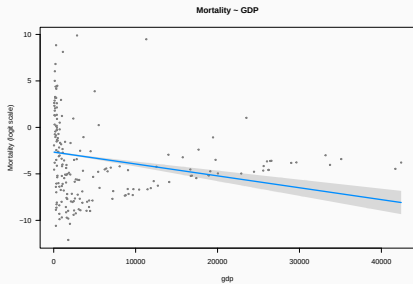
# Trying polynomial predictor (GDP + GDP<sup>2</sup>)



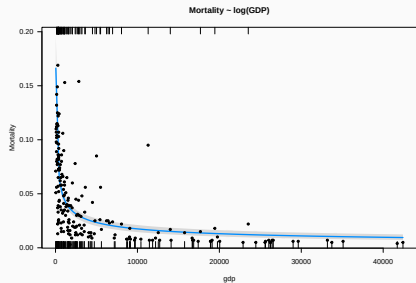
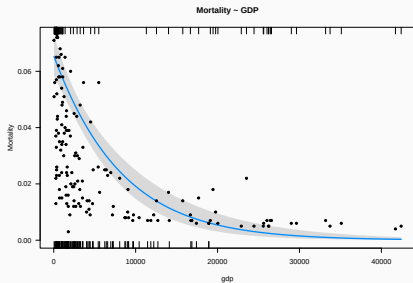
# Think about the shape of relationships



# Trying log(GDP)



# Trying $\log(\text{GDP})$



- `moth.csv`: Probability of moth predation on trunk trees depending on morph (light/dark) and distance to Liverpool ([Bishop 1972](#))

- `moth.csv`: Probability of moth predation on trunk trees depending on morph (light/dark) and distance to Liverpool ([Bishop 1972](#))
- `seedset.csv`: Comparing seed set among plants (Data from [Harder et al. 2011](#))

- **moth.csv**: Probability of moth predation on trunk trees depending on morph (light/dark) and distance to Liverpool ([Bishop 1972](#))
- **seedset.csv**: Comparing seed set among plants (Data from [Harder et al. 2011](#))
- **soccer.csv**: Probability of scoring penalty depending on goalkeeper's team being ahead, behind or tied ([Roskes et al 2011](#))

## Moth predation

---

## The industrial revolution and evolution of dark morphs



```
moth <- read.csv('data/moth.csv')
```

|   | MORPH | DISTANCE | PLACED | REMOVED |
|---|-------|----------|--------|---------|
| 1 | light | 0.0      | 56     | 17      |
| 2 | dark  | 0.0      | 56     | 14      |
| 3 | light | 7.2      | 80     | 28      |
| 4 | dark  | 7.2      | 80     | 20      |
| 5 | light | 24.1     | 52     | 18      |
| 6 | dark  | 24.1     | 52     | 22      |

## Creating new variable: REMAIN

```
moth$REMAIN <- moth$PLACED - moth$REMOVED
```

|   | MORPH | DISTANCE | PLACED | REMOVED | REMAIN |
|---|-------|----------|--------|---------|--------|
| 1 | light | 0.0      | 56     | 17      | 39     |
| 2 | dark  | 0.0      | 56     | 14      | 42     |
| 3 | light | 7.2      | 80     | 28      | 52     |
| 4 | dark  | 7.2      | 80     | 20      | 60     |
| 5 | light | 24.1     | 52     | 18      | 34     |
| 6 | dark  | 24.1     | 52     | 22      | 30     |

## Did some morph have higher predation overall?

Call:

```
glm(formula = cbind(REMOVED, REMAIN) ~ MORPH, family = binomial,  
     data = moth)
```

Coefficients:

|             | Estimate | Std. Error | z value | Pr(> z )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | -0.57752 | 0.09473    | -6.097  | 1.08e-09 *** |
| MORPHlight  | -0.40331 | 0.13925    | -2.896  | 0.00377 **   |

---

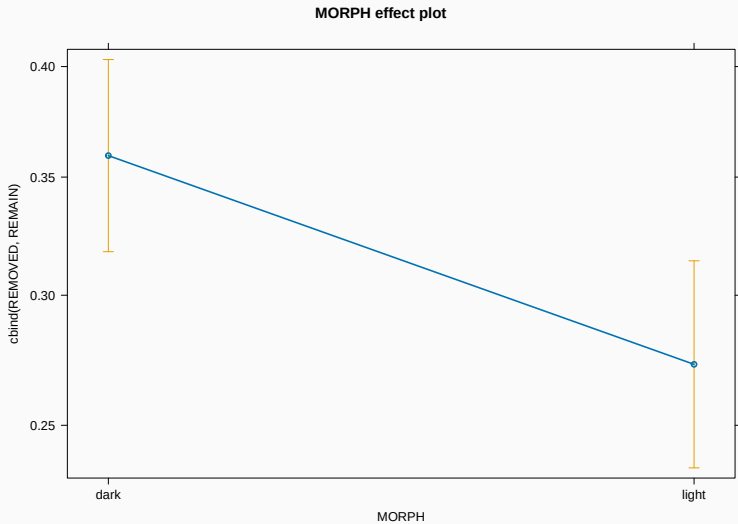
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 35.385 on 13 degrees of freedom  
Residual deviance: 26.936 on 12 degrees of freedom  
AIC: 93.61

Number of Fisher Scoring iterations: 4

## Did some morph have higher predation overall?



## Did predation increase farther from city centre?

Call:

```
glm(formula = cbind(REMOVED, REMAIN) ~ DISTANCE, family = binomial,  
     data = moth)
```

Coefficients:

|             | Estimate  | Std. Error | z value | Pr(> z )     |
|-------------|-----------|------------|---------|--------------|
| (Intercept) | -0.925861 | 0.136634   | -6.776  | 1.23e-11 *** |
| DISTANCE    | 0.005268  | 0.003984   | 1.322   | 0.186        |

---

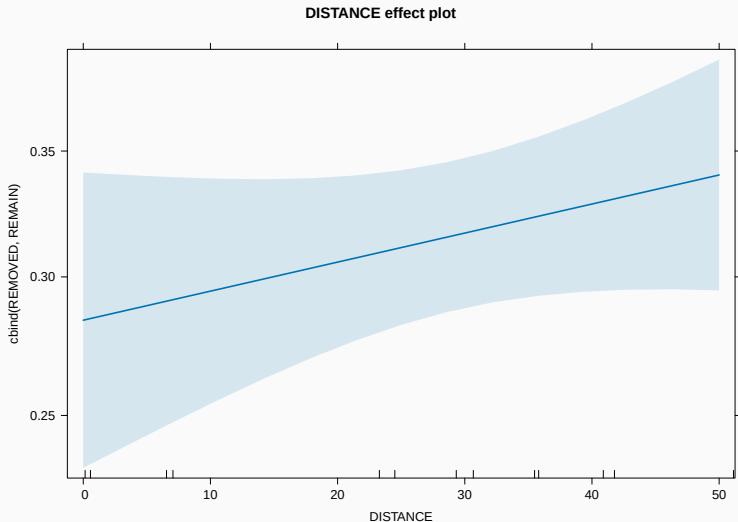
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 35.385 on 13 degrees of freedom  
Residual deviance: 33.626 on 12 degrees of freedom  
AIC: 100.3

Number of Fisher Scoring iterations: 4

## Did predation increase farther from city centre?



# Did dark morph have lower predation in city & light have lower predation in countryside?

Call:

```
glm(formula = cbind(REMOVED, REMAIN) ~ MORPH * DISTANCE, family = binomial,  
     data = moth)
```

Coefficients:

|                     | Estimate  | Std. Error | z value | Pr(> z ) |     |
|---------------------|-----------|------------|---------|----------|-----|
| (Intercept)         | -1.128987 | 0.197906   | -5.705  | 1.17e-08 | *** |
| MORPHlight          | 0.411257  | 0.274490   | 1.498   | 0.134066 |     |
| DISTANCE            | 0.018502  | 0.005645   | 3.277   | 0.001048 | **  |
| MORPHlight:DISTANCE | -0.027789 | 0.008085   | -3.437  | 0.000588 | *** |

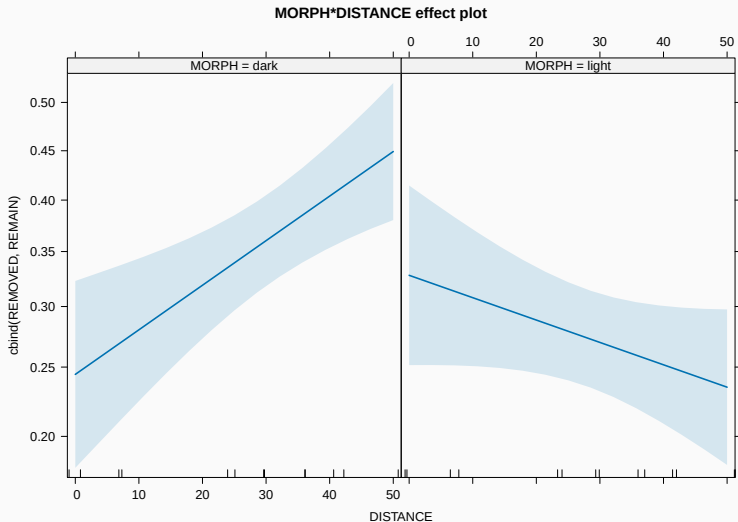
---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

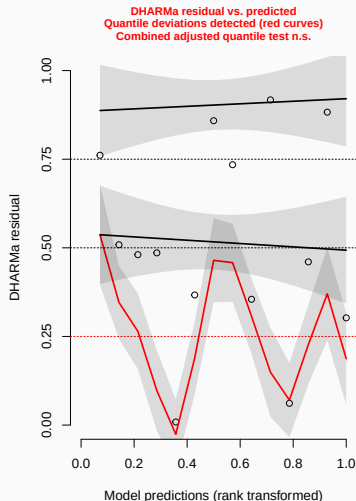
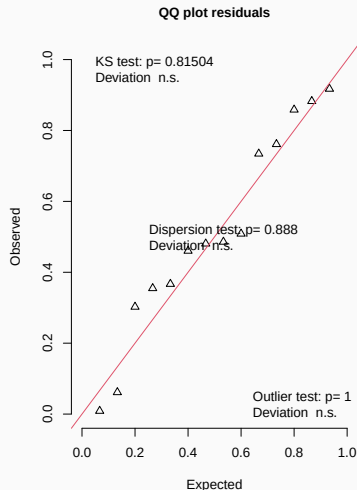
Null deviance: 35.385 on 13 degrees of freedom  
Residual deviance: 13.230 on 10 degrees of freedom  
AIC: 83.904

Did dark morph have lower predation in city & light have lower predation in countryside?



```
simulateResiduals(pred.int, plot = TRUE)
```

DHARMa residual



## Seed set among plants

---

## Seed set among plants



## Seed set among plants

```
# A tibble: 6 x 6
```

|   | species    | plant | pcmass | fertilized | seeds | ovulecnt |
|---|------------|-------|--------|------------|-------|----------|
|   | <chr>      | <dbl> | <dbl>  | <dbl>      | <dbl> | <dbl>    |
| 1 | ferruginea | 2     | 0      | 70         | 52    | 330      |
| 2 | ferruginea | 2     | 0.2    | 321        | 188   | 461      |
| 3 | ferruginea | 2     | 0.485  | 351        | 278   | 435      |
| 4 | ferruginea | 2     | 0.737  | 386        | 301   | 430      |
| 5 | ferruginea | 2     | 1      | 367        | 342   | 419      |
| 6 | ferruginea | 3     | 0      | 185        | 39    | 470      |

## Questions:

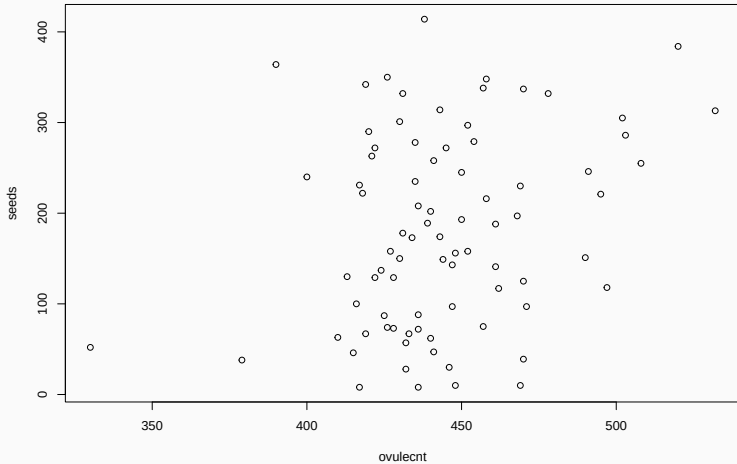
<https://pollev.com/franciscorod726>

- Is seed set related to proportion of outcross pollen (pcmass)?

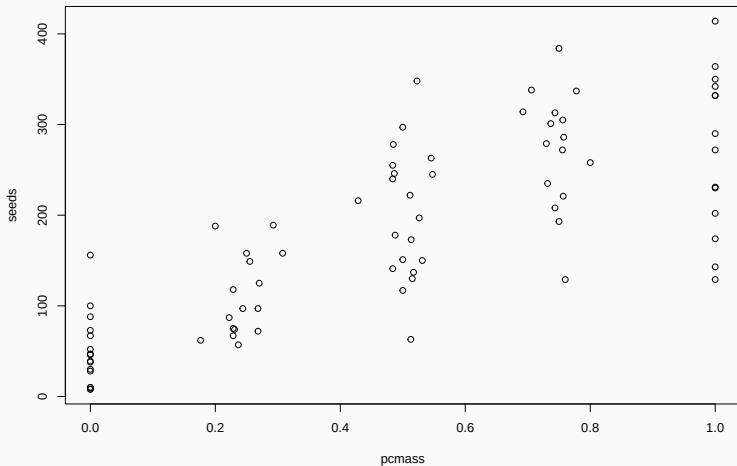
<https://pollev.com/franciscorod726>

- Is seed set related to proportion of outcross pollen (pcmass)?
- Which plant had lower seed set?

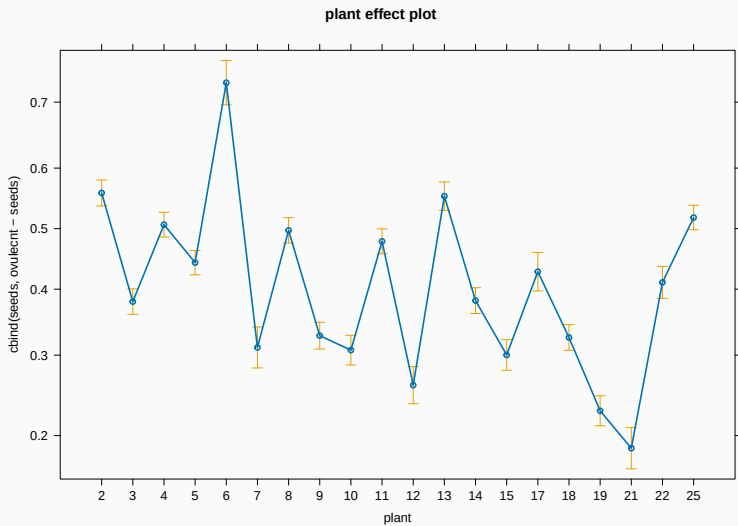
## Number of seeds vs Number of ovules



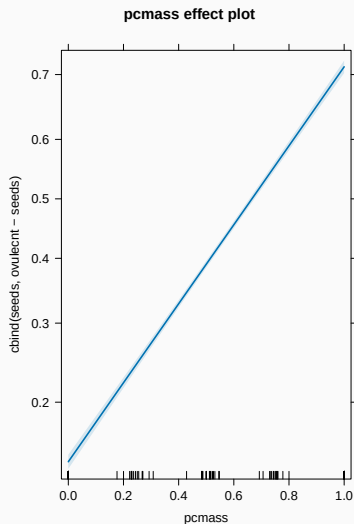
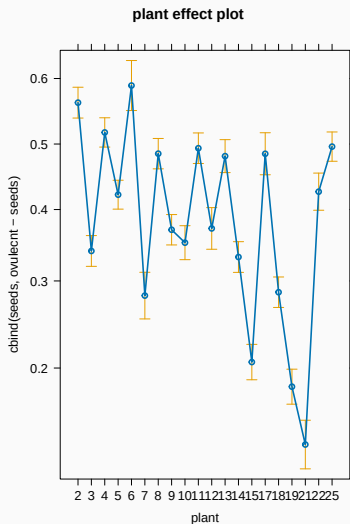
## Number of seeds vs Proportion outcross pollen



## Seed set across plants



## Seed set ~ outcross pollen



## Probability of scoring penalty

---

```
soccer <- read.csv('data/soccer.csv')  
soccer
```

|   | GoalkeeperTeam | Nshots | Scored |
|---|----------------|--------|--------|
| 1 | Behind         | 20     | 18     |
| 2 | Tied           | 90     | 71     |
| 3 | Ahead          | 75     | 55     |

Does probability of scoring penalty depends on match situation?

<https://pollev.com/franciscorod726>

# Probability of scoring depending on match situation

